# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1 ANN10** | **base10k** | 0.0321 | 0.142 | 1.26 | 1.01 | 0.17 | 0.99 | 0.0814 | 1.02 | 1.21 | 0 | 18/18 |
| **1 ANN10** | **soft10k** | 0.67 | 1.55 | 1 | 66.6 | 4.22 | 4.11 | 1 | 0.999 | 704 | 0 | 18/18 |
| **1 ANN10** | **long** | 0.635 | 1.21 | 1 | 1 | 2.85 | 24.7 | 1 | 1 | 14.7 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2 P-only DeepONet10** | **base10k** | 0.0578 | 0.114 | 0.951 | 0.994 | 0.165 | 0.993 | 0.123 | 0.995 | 1.19 | 0 | 18/18 |
| **2 P-only DeepONet10** | **soft10k** | 0.555 | 1.34 | 1 | 1 | 2.77 | 1 | 1 | 1.35e+04 | 1 | 0 | 18/18 |
| **2 P-only DeepONet10** | **long** | 0.553 | 1.37 | 1 | 1 | 3.14 | 1 | 1 | 1.24e+04 | 1 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **3 ANN10 + P-only DeepONet10** | **base10k** | 0.0373 | 0.0777 | 0.901 | 0.92 | 0.244 | 0.993 | 0.0941 | 0.995 | 1 | 0 | 18/18 |
| **3 ANN10 + P-only DeepONet10** | **soft10k** | 0.542 | 1.64 | 1 | 1 | 5 | 1 | 1 | 0.998 | 815 | 0 | 18/18 |
| **3 ANN10 + P-only DeepONet10** | **long** | 0.58 | 1.46 | 1 | 0.999 | 4.07 | 11.9 | 1 | 0.999 | 17.2 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **4 T/P DeepONet10** | **base10k** | 0.0466 | 0.138 | 1.02 | 0.786 | 0.519 | 0.998 | 0.102 | 1.15 | 1.35 | 0 | 17/18 |
| **4 T/P DeepONet10** | **soft10k** | 0.553 | 1.56 | 1 | 1 | 4.96 | 1 | 1 | 627 | 1 | 0 | 18/18 |
| **4 T/P DeepONet10** | **long** | 0.545 | 1.8 | 1 | 1 | 5.09 | 1 | 1 | 0.983 | 1.81 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **5 ANN10 + T/P DeepONet10** | **base10k** | 0.0491 | 0.0957 | 0.962 | 0.82 | 0.174 | 0.97 | 0.066 | 0.921 | 0.993 | 0 | 18/18 |
| **5 ANN10 + T/P DeepONet10** | **soft10k** | 0.542 | 1.64 | 1 | 1 | 5 | 1 | 1 | 0.998 | 761 | 0 | 18/18 |
| **5 ANN10 + T/P DeepONet10** | **long** | 0.544 | 1.64 | 1 | 1 | 5.01 | 1 | 1 | 0.999 | 8.34 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within T/P/$\phi$ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **6 T/P/$\phi$ DeepONet10** | **base10k** | 0.0835 | 0.0903 | 1 | 1 | 0.345 | 1 | 0.134 | 1.41 | 1.92 | 0 | 17/18 |
| **6 T/P/$\phi$ DeepONet10** | **soft10k** | 0.533 | 1.47 | 1 | 1 | 4.94 | 1 | 1 | 4.69 | 1 | 0 | 18/18 |
| **6 T/P/$\phi$ DeepONet10** | **long** | 0.542 | 1.48 | 1 | 1 | 4.94 | 1 | 1 | 0.987 | 2.91 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

# Lambda 1000 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P/$\phi$ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

| Model / same-state mean | Stage | $T$ | $Y_{H_2}$ | $Y_H$ | $Y_O$ | $Y_{O_2}$ | $Y_{OH}$ | $Y_{H_2O}$ | $Y_{HO_2}$ | $Y_{H_2O_2}$ | $Y_{N_2}$ | Complete ICs |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **base10k** | 0.0633 | 0.0986 | 1 | 0.584 | 0.195 | 0.998 | 0.192 | 0.998 | 1 | 0 | 18/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **soft10k** | 0.544 | 1.64 | 1 | 1 | 4.99 | 1 | 1 | 0.997 | 830 | 0 | 18/18 |
| **7 ANN10 + T/P/$\phi$ DeepONet10** | **long** | 0.543 | 1.64 | 1 | 0.999 | 5.01 | 1 | 1 | 1 | 3.85 | 0 | 18/18 |

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.